# Sorting

CSE 101 Section 03

Introduction to Algorithms and Computing

Fall 2022

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# The sorting problem

- Putting the elements of a list into an order.
  - Ascending
  - Descending

- Example:
  - Sort Amazon search results by:
    - Number of stars
    - Popularity
    - Price

# The sorting algorithm

- A sorting algorithm is used to arrange elements of an array/list in a specific order.
- For example

# The sorting algorithm

- In the previous example, we are sorting the array in ascending order.

- There are various sorting algorithms that can be used to complete this operation (we will cover some later).

- Some sorting algorithms are better than others, in terms of how long they take (i.e., time complexity), and how much memory space they need (i.e., space complexity).

# Exercise

- As a team of 6 students, do the following exercise:
  - Write a Matlab function to sort an array of any size.

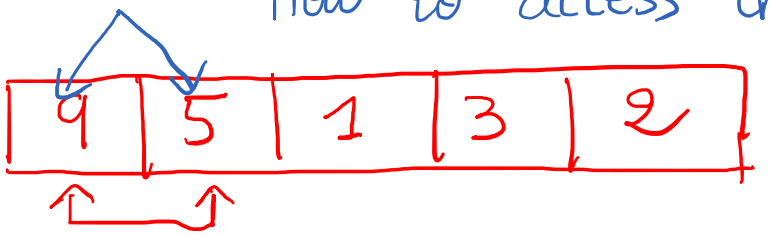
```
anArray = [9 5 4 3];
mysortingalgorithm(anArray);

function mysortedarray = mysortingalgorithm(myarray)
    % Your task: code to sort the array
end
```

# Exercise

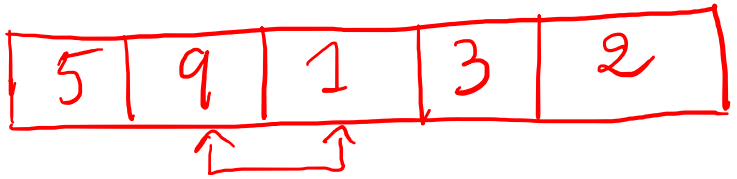
- STEP 1a: Sketch on paper your thought process,
  - How would you move the elements around to get the list in order?
  - Do you need a loop? If yes, which one?
  - What do you need to compare? What do you need to keep track of?
  - Try to draw what a random array would look like after each iteration and/or swap.

How to access these 2 elements to compare them?

Step 1




Step 2



# Exercise

- STEP 1b: Write the steps of your solution in plain english

  Something like:

  *For each element in my array*

  *Do this*

  *Then do that*

# Exercise

- STEP 2: Convert your algorithm in plain English to Matlab by filling in the following function

```
anArray = [9 5 4 3];
mysortingalgorithm(anArray);

function mysortedarray = mysortingalgorithm(myarray)
     % Your task: code to sort the array
end
```